



# Welcome to QuACK!

Week 4

10/1/2026

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# Structure of the course

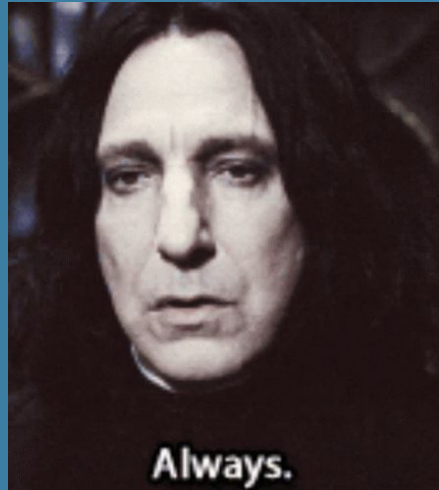
|         |       |                                                    |
|---------|-------|----------------------------------------------------|
| Week 1  | 9/10  | Intro to R & Programming                           |
| Week 2  | 9/17  | Intro to the tidyverse and data wrangling - Part 1 |
| Week 3  | 9/24  | Intro to the tidyverse and data wrangling - Part 2 |
| Week 4  | 10/1  | Intro to data visualization (ggplot) - Part 1      |
| Week 5  | 10/8  | Intro to data visualization (ggplot) - Part 2      |
| Week 6  | 10/15 | Error Messages/Review/Q&A                          |
| Week 7  | 10/22 | Intro to Basic Stats - Part 1                      |
| Week 8  | 10/29 | Intro to Basic Stats - Part 2                      |
| Week 9  | 11/5  | Putting it all together                            |
| Week 10 | 11/12 | <b>NO CLASS</b>                                    |
| Week 11 | 11/19 | Open-Science, Github & Git                         |

# Today's Agenda: Plotting

- Intro to ggplot2
- Demo: data visualization
- Practice



# When should you visualize your data?

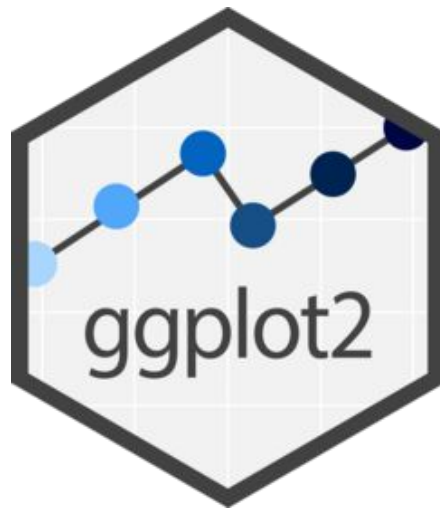


# Ok...but why?

- Helps to easily identify trends in your data – maybe some you expected, and others you didn't expect
- Helps find outliers in your data – these may be throwing off your stats without you even knowing!
- In some cases, it can help you decide which statistical tests are appropriate for the kind of data you have
- Can help you easily communicate your findings to collaborators and the broader scientific community

# Introducing ggplot2

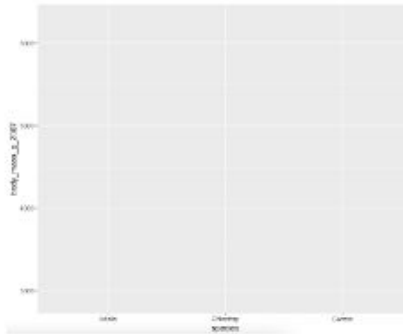
- Last week we learned about Tidyverse. We used some Tidyverse packages like `readr` and `dplyr` to clean our data
- Today we are learning about another package in Tidyverse called `ggplot2` that is used for data visualization
- Why ggplot?
  - It's easy to use
  - It works well with the rest of Tidyverse (and R in general)
  - It creates very flexible and easily customizable plots



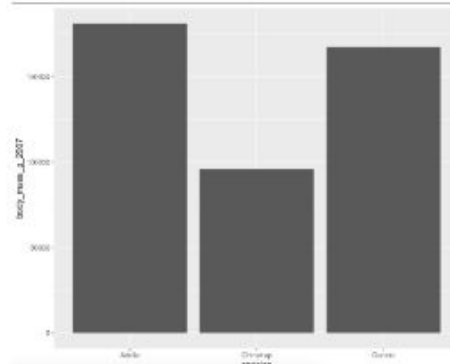
# Build any plot with ggplot

- You can combine layers in ggplot2 to create a plot
- The way ggplot2 is so flexible is by offering many different layers, each with many different customization options

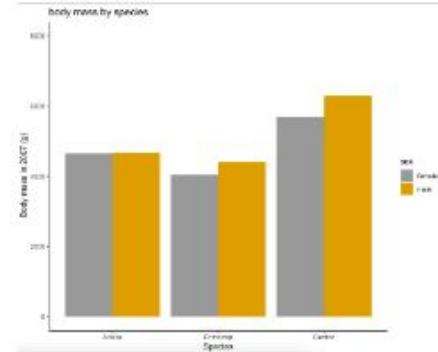
Step 1: create a plot and designate variables



Step 2: Decide how to represent the data



Step 3: customize



# ggplot2 terminology

- **Creating a ggplot object (this is your “canvas”)**

```
ggplot(data = myDataFrame, aes(x = x_series, y = y_series))
```

- **Adding geometric objects to your plot. These tell ggplot how to represent your data.**

```
ggplot(data = myDataFrame, aes(x = x_series, y = y_series)) +
```

```
  geom_col()
```

Note that ggplot layers are connected by “+” not “%>%”

- **Add other layout and design features**

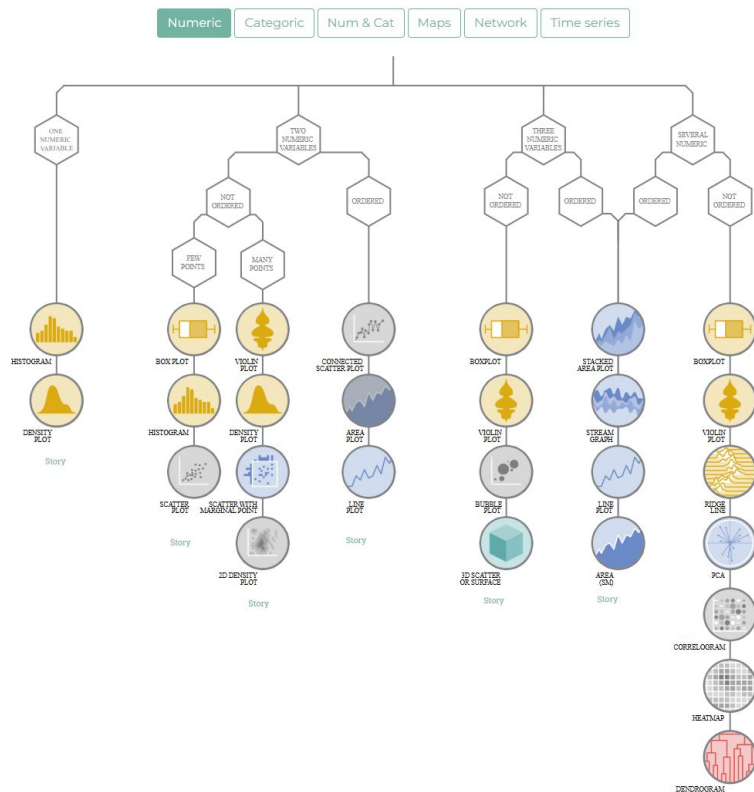
```
ggplot(data = myDataFrame, aes(x = x_series, y = y_series)) +
```

```
  geom_col() +
```

```
  theme_classic()
```

# And lastly, a helpful dataviz resource:

<https://www.data-to-viz.com/>



# Time to get started in R!

- Go to <https://tinyurl.com/quack-fall2026>
- Download and unzip the course materials
- You will see many files:
  - Two R Markdown (.rmd) files
  - A .csv file
  - Two PDF files – slides and a practice doc
  - Some other folders – ignore these for now
- Open the R Markdown file marked “starter\_code” – this is where we will code together for the first part of the session
- The .csv file contains the data we will be working with
- The practice PDF contains practice exercises for you to do during the second part of the session



# Exit Ticket

Link: <https://tinyurl.com/quack-fall2026-exit-survey>

